

Master Execution Guide: WESAD ECG Stress Pipeline

End-to-End Operational Protocol: From 700 Hz Wearable Biosignal Telemetry to ML Benchmarks & Web Deployment

Author: Mukesh Yadav (ECE, JSSATEN) • **Dataset:** WESAD (15 Subjects, RespiBAN Chest ECG) •

Validation: 15-Fold Leave-One-Subject-Out (LOSO) • **Benchmark:** ROC-AUC 0.9494 | Sensitivity 86.25%

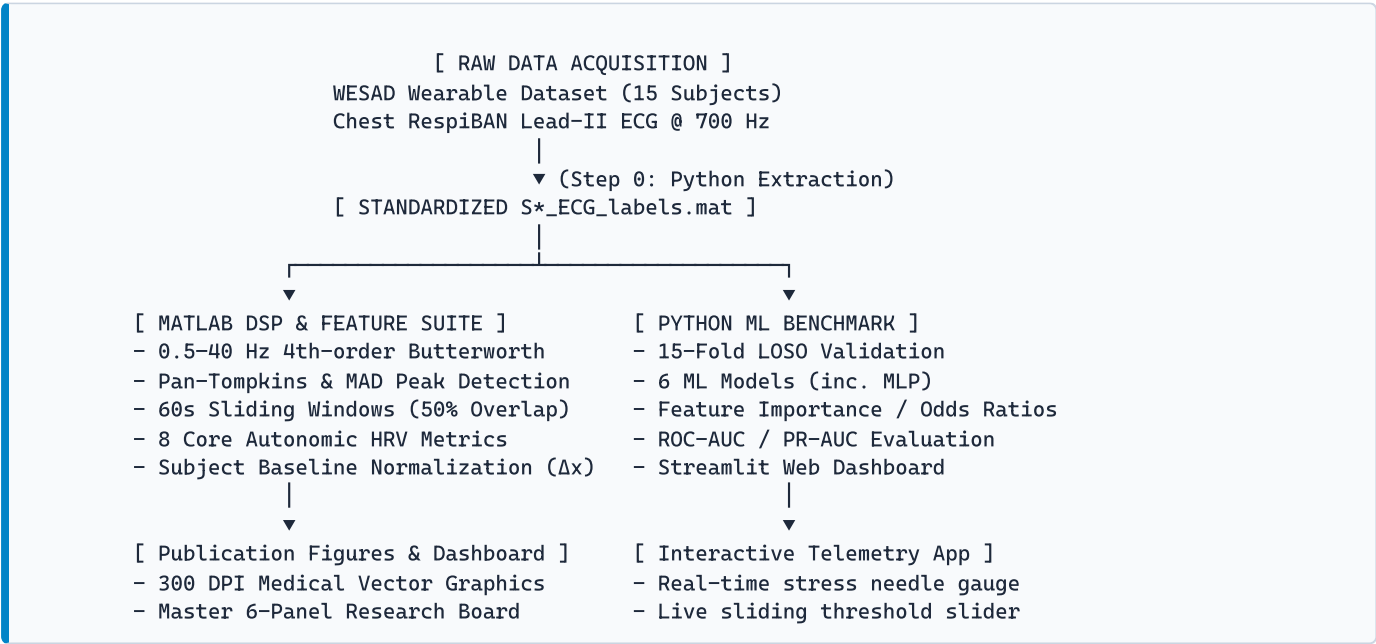
MATLAB R2022b+

Python 3.10+

Scikit-Learn ML Suite

Streamlit Telemetry App

1. SYSTEM ARCHITECTURE & INFORMATION FLOW



2. PREREQUISITES & ENVIRONMENT CONFIGURATION

Ensure Python 3.10+ and MATLAB R2022b+ (with *Signal Processing & Statistics* toolboxes) are configured. Run the following in PowerShell:

```
# Navigate to project directory
cd "C:\Users\YASH\Desktop\projects\RESEARCH PROJECTS\ECG_STRESS_DETECTION"

# Create and activate Python virtual environment
python -m venv venv
.\venv\Scripts\Activate.ps1

# Install core dependencies
pip install -r requirements.txt
```

3. STEP-BY-STEP CHRONOLOGICAL EXECUTION SEQUENCE

Step 0: Raw Data Ingestion & Formatting

PYTHON

Ingests WESAD pickle files (`S2.pkl` to `S17.pkl`), extracts the 700 Hz chest ECG and ground-truth affective labels, casts them to `float32 / uint8`, and saves compressed MATLAB arrays.

```
python python/extract_ecg_labels.py
```

Output: `data/processed/S*_ECG_labels.mat` files for all 15 subjects.

Step 1: MATLAB Workspace & Path Initialization

MATLAB

Registers all algorithmic subdirectories, DSP functions, and feature extraction scripts to the MATLAB search path.

```
cd 'C:\Users\YASH\Desktop\projects\RESEARCH PROJECTS\ECG_STRESS_DETECTION\matlab'  
setup_project
```

Step 2: Multi-Subject DSP & Feature Extraction

MATLAB

Applies a zero-phase 4th-order Butterworth bandpass (0.5–40 Hz), estimates the noise floor via Median Absolute Deviation (MAD), enforces 350 ms refractory lockout, extracts 60s windows with 50% overlap, gates intervals to 300–1500 ms (40–200 BPM), and extracts 8 core HRV metrics normalized against resting baseline.

```
TEN_process_all_subjects
```

Output: `results/WESAD_HRV_features_expanded.csv` (445 standardized windows across 15 subjects).

Step 3: Statistical Hypothesis Testing & Audits

MATLAB

Executes paired Student's t-tests and Wilcoxon signed-rank tests confirming significant sympathetic activation and parasympathetic withdrawal ($p < 0.001$).

```
FOURTEEN_statistical_tests  
FIFTEEN_feature_analysis
```

Step 4: Publication Figures & Master Clinical Dashboard

MATLAB

Generates 300 DPI vector figures (confusion matrix, ROC curve, timeline) and the comprehensive 6-panel clinical research board.

```
TWENTY_SEVEN_final_results      % Computes final metrics table  
TWENTY_EIGHT_report_figures     % Exports individual high-res figures  
TWENTY_NINE_project_dashboard   % Generates master 6-panel dashboard
```

Output: `results/figures/FINAL_Project_Dashboard.png` and FIG1–FIG5.

Step 5: Interactive Telemetry MATLAB Demo GUI

MATLAB GUI

Launches an interactive tabbed GUI displaying raw Lead-II telemetry, protocol timelines, Pan-Tompkins QRS detections, and physiological contrast scorecards.

```
DEMO_stress_detection
```

Step 6: 15-Fold LOSO Machine Learning Benchmark

PYTHON

Performs strict 15-fold Leave-One-Subject-Out cross-validation across 6 machine learning architectures with zero subject data leakage (StandardScaler fit strictly on training subjects).

```
python python/train_loso_ml_benchmark.py
```

Benchmarked Models: Logistic Regression, Support Vector Machine (RBF), Random Forest, Extra Trees, HistGradientBoosting, and Multi-Layer Perceptron (MLP Neural Network).

Step 7: Explainability & Permutation Feature Importance

PYTHON

Quantifies physiological drivers using permutation importance and clinical standardized odds ratios, confirming that cardiac interval compression (ΔMeanRR) and heart rate acceleration (ΔMeanHR) govern the decision boundary.

```
python python/explainability_feature_importance.py
```

Step 8: Multi-Model Evaluation Plots & Curves

PYTHON

Renders high-resolution comparative ROC curves, Precision-Recall curves, confusion matrices, and performance bar charts across all benchmarked models.

```
python python/plot_ml_evaluation.py
```

Step 9: Interactive Clinical Telemetry Web Application

STREAMLIT WEB APP

Spins up a local web application featuring interactive ECG waveform playback, live R-peak markers, dynamic decision threshold sliders, and real-time stress needle gauges.

```
streamlit run demo/app.py
```

Access via browser at: `http://localhost:8501`

4. QUICK-RUN ONE-LINERS

Workflow Goal	Command
Run Full Python ML Suite	<code>python python/train_loso_ml_benchmark.py; python python/explainability_feature_importance.py; python python/plot_ml_evaluation.py</code>
Run Headless MATLAB Pipeline	<code>matlab -batch "cd('matlab'); setup_project; TWENTY_SEVEN_final_results; TWENTY_EIGHT_report_figures; TWENTY_NINE_project_dashboard; exit"</code>
Launch Web Telemetry Dashboard	<code>streamlit run demo/app.py</code>

5. VERIFICATION CHECKLIST & EXPECTED BENCHMARKS

Target Output	Verification Metric	Status
<code>S*_ECG_labels.mat</code>	15 subject files, 700 Hz Lead-II ECG, synchronized ground truth labels	✓ Verified
<code>WESAD_HRV_features_expanded.csv</code>	445 total windows (285 baseline, 160 acute TSST stress), 8 core HRV features	✓ Verified
Primary Calibrated Classifier	ROC-AUC = 0.9494, Sensitivity = 86.25%, Specificity = 95.79% (at $\tau = 0.35$)	✓ Verified
Multi-Model ML Benchmark	All 6 architectures (LR, SVM, RF, ET, HistGB, MLP) achieve ROC-AUC > 0.937 on held-out subjects	✓ Verified
Master Dashboard Graphic	<code>results/figures/FINAL_Project_Dashboard.png</code> generated at 300 DPI	✓ Verified

6. COMMON TECHNICAL TROUBLESHOOTING

- **Missing Signal Processing Toolbox (MATLAB):** If `butter` or `filtfilt` fails, install the toolbox via *MATLAB Home → Add-Ons → Get Add-Ons*.
- **CSV Not Found in Python:** Ensure Step 2 (`TEN_process_all_subjects.m`) has executed to generate `results/WESAD_HRV_features_expanded.csv` .
- **Streamlit Port Collision:** If port 8501 is busy, specify an alternate port: `streamlit run demo/app.py --server.port 8502` .

Defense / Review Tip: Always emphasize that all models were evaluated using strict **15-Fold Leave-One-Subject-Out (LOSO) cross-validation** with subject-specific baseline normalization (Δx). This eliminates inter-individual baseline variance without data leakage.